

Journal 01: Operating System Security and Access Control

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Many of us put faith in our devices to protect personal information, from family photos and documents to health data and social security numbers. The operating system (OS) plays a crucial role in managing security and providing data integrity. The three main desktop OSs—Windows, macOS, and Linux—approach security in distinct ways. First, the security roles and responsibilities of the OS will be enumerated. Next, the security architectures of the three main OSs will be compared as to how they ensure confidentiality, integrity, and availability through access control, encryption, and authentication. Finally, the key differences and similarities among the three approaches will be summarized.

In its role as the intermediary between the user and the hardware, the OS is responsible for providing both protection and security. A protection is a mechanism for access to resources, while security refers to the defense of a system from targeted attacks (Silberschatz et al., 2023). The protection of resources is mainly provided through access control, in which only authorized users or processes may access a resource. Most often, access control utilizes user and group identifiers (Silberschatz et al., 2023). Security is often realized through a combination of additional software shipped with the OS and trusted hardware protections such as kernel mode (Adekotujo et al., 2020). While the OS plays a large part in protecting computer systems, it is still up to system users to ensure proper configuration.

Building on these core responsibilities, each of the three main OSs takes a different approach to security. Windows uses a centralized architecture, relying on system-wide credential storage and policy management through the Security Accounts Manager or Active Directory (Albatli et al., 2023). Linux opts for a more decentralized and modular approach with an emphasis on configurability (Adekotujo et al., 2020). macOS utilizes a layered architecture, with a custom microkernel for hardware control and a UNIX-based access control scheme (Albatli et

al., 2023). Additionally, macOS utilizes sandboxing to ensure applications remain isolated (Silberschatz et al., 2023).

Despite their architectural differences, all three operating systems rely on a common set of security mechanisms. Access control, authentication, and encryption are key components in ensuring confidentiality, integrity, and availability. Windows implements access control using access control lists (ACLs) and role-based permissions, while Linux and macOS opt for discretionary access control (DAC) on a file-by-file basis (Adekotujo et al., 2020). The SELinux kernel module provides support for mandatory access control (MAC) needed in more centralized enterprise environments (Albatli et al., 2023). All three OSs support typical password authentication. Linux offers the most configurability through pluggable authentication modules (PAMs), which add functionality for alternative authentication form factors. BitLocker provides full disk encryption for Windows, relying on the trusted platform module (TPM) as a hardware root-of-trust (Microsoft, n.d.). FileVault is the macOS equivalent of BitLocker (Adekotujo et al., 2020). Consistent with its modular design, Linux offers multiple encryption options that require manual configuration (Albatli et al., 2023).

These mechanisms directly support the three core security goals known as the confidentiality, integrity, and availability (CIA) triad. Confidentiality ensures that only authorized users may view specific content or resources, primarily through access control and encryption. Integrity ensures that only authorized users may modify specific content or resources. Like confidentiality, integrity relies on access control, with the addition of mechanisms such as checksums and version control (Adekotujo et al., 2020). Availability ensures that all of a computer's resources remain accessible and undamaged. Availability is secured through redundancy and strong antivirus protection. Windows offers Windows Defender

as its main antivirus software, in combination with a secure boot process to protect against rootkits (Microsoft, n.d.). Various open-source antivirus alternatives are available for Linux, while macOS implements GateKeeper and XProtect to defend against malware (Adekotujo et al., 2020).

While these principles are implemented across all three operating systems, real-world security outcomes vary significantly. Windows likely faces the toughest security challenge due to its large user base (Albatli et al., 2023). Threat actors gain more targets, including high-value enterprise users, by focusing on Windows. Additionally, Windows and macOS tend to be slower in rolling out security fixes due to their corporate structure, while the Linux community is able to rapidly respond to newly discovered vulnerabilities (Albatli et al., 2023). However, Linux relies on proper configuration. Modules and applications are often left with default settings, effectively nullifying their security benefits. Windows and Linux both suffer from poor defaults that allow administrator or root access, violating the principle of least privilege (Silberschatz et al., 2023). Arguably, macOS provides the most sensible security defaults.

In summary, a computer's OS plays a vital role in protecting system resources. Access control, authentication, and encryption work together to ensure the CIA triad. However, Windows, macOS, and Linux accomplish these tasks in slightly different ways. Windows utilizes a centralized approach with ACLs and software security applications such as BitLocker and Windows Defender. Linux opts for a decentralized and highly configurable architecture using DAC and a large library of kernel modules. Combining aspects of both sides, macOS follows a layered approach, mixing DAC with a custom microkernel and sandboxing to isolate applications. Regardless of the OS, a computer system is never fully secure. Over time, new vulnerabilities are discovered, patches are made, and the cycle begins again.

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